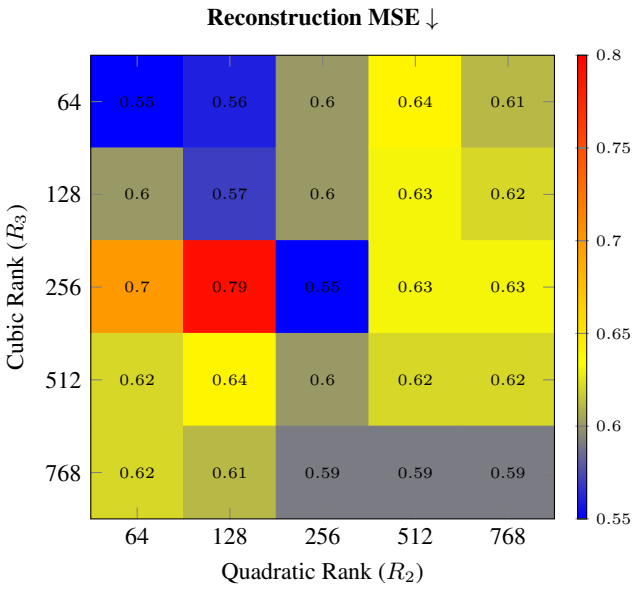

Figure 4. Reconstruction MSE for different R_2 and R_3 values, with $R_1 = 768$, using activations from GPT-2 Small.

outperforms standard SAEs across all sparsity levels, with the gap widening at higher k. PolySAE also exhibits lower variance across widths, enabling competitive performance with fewer active features in smaller dictionaries where standard SAEs would require substantially more.

Semantic Concentration Across Features. We next ask *(Q4) whether PolySAE concentrates semantic signal into fewer features*. Table 2 reports the F1 gain $\Delta_{1\text{–}5}$ when expanding from K=1 to K=5 active features, averaged across all probing datasets and sparsifiers. PolySAE exhibits smaller gains than standard SAEs in 3 out of 4 models, with the largest difference on GPT-2 Small (-7.6%). One interpretation is that PolySAE concentrates semantic information into fewer features, reducing the marginal benefit of additional features. This indicates that higher-order interactions absorb contextual variability while PolySAE's linear features remain more semantically focused.

Ablating different ranks. Figure 4 examines the effect of interaction ranks on reconstruction, for fixed $R_1 = 768$ on GPT-2 Small. PolySAE achieves competitive reconstruction with modest interaction ranks ($R_2 = R_3 = 64$). Increasing ranks beyond this does not improve reconstruction, suggesting that the additional capacity is unnecessary for capturing interaction structure in this setting.

5. Qualitative Analysis: Making Sense of Learned Interactions

To better understand the learnt interactions, we firstly ask *(Q5) whether PolySAE's higher-order terms encode genuine compositional structure or merely reflect surface-level co-occurrence*. To study this, we analyze SAE and PolySAE activations trained with Top-K sparsification on GPT-2 small, using 1M OpenWebText texts. For each feature pair (i, j), we first define the learned quadratic interaction strength $B_{ij} = \lambda_2 \left\|(\boldsymbol{u}_i \odot \boldsymbol{u}_j)^\top \boldsymbol{C}^{(2)}\right\|_2$, which depends only on the trained decoder parameters and measures how much representational capacity the model allocates to the (i, j) interaction. Second, we compute the empirical co-occurrence frequency N_{ij} by counting token positions in which both features appear in the top-K active set across the same corpus. If polynomial interactions merely replicated bigram statistics, B_{ij} and N_{ij} would correlate strongly. As a baseline, we consider the empirical covariance of SAE activations, which captures the full pairwise structure accessible to a linear model.

Table 2. Mean F1 Gain from K=1 to K=5 after averaging the results from all 6 probing datasets and all 3 sparsifiers.

Model	SAE	PolySAE	PolySAE effect
GPT-2 Small	+14.3	**+6.7**	–7.6
Pythia-410m	+11.0	**+10.5**	–0.5
Pythia-1.4b	**+10.3**	+10.5	+0.2
Gemma2-2b	+13.6	**+10.6**	–3.0
Overall	+12.3	**+9.6**	–2.7

As expected, this covariance correlates strongly with co-occurrence frequency ($r = 0.82$). In contrast, PolySAE's learned interactions exhibit negligible correlation with co-occurrence ($r = 0.06$), indicating that interaction capacity is allocated based on criteria largely orthogonal to frequency.

Since higher-order dictionaries do not simply encode co-occurrence, we finally ask *(Q6) whether the learned interactions are interpretable*. To analyze this structure, we construct a dictionary mapping each feature to its most activating tokens, then examine feature pairs and triples with high interaction strength by extracting representative contexts in which the corresponding features co-activate.

Selected examples in Table 3 and Table 4 illustrate the qualitative structure captured by PolySAE's higher-order terms. Second-order interactions often correspond to coherent phrase-level compositions that are not recoverable from either feature in isolation, such as *coffee* × *star* yielding contexts referring to *Starbucks*, a highly non-linear semantic mapping. In contrast, SAEs typically activate broad or weakly related features in these contexts, failing to recover the composed meaning. Third-order interactions further refine such compositions by conditioning on additional context. For example, PolySAE distinguishes financial *investing* from unrelated `-ing` usages by integrating morphological cues with market-related features, and disambiguates

Table 3. **Second-Order Interaction Examples Captured by PolySAE.** Quadratic interactions bind two features to capture context-dependent semantic structure beyond co-occurrence. SAE often recovers individual components but fails to represent the composed meaning.

Poly F_1	**Poly** F_2	**Context**	**SAE**	**Observed Pattern**
[star, stars]	[coffee, tea]	We've all certainly heard of beers brewed with espresso, but how about one with an espresso shot poured over the top? **Starbucks**	[Apple, Google]	*The interaction binds features to represent a specific named entity creating a new semantic category.*
[surgery, repair]	[Trans, LGBT]	Some in the transgender community are worried a suspicious fire at a Montreal clinic will add delays to an already lengthy process to get gender reassignment **surgery**	[birth, baby]	*Specialization: a general concept (surgery) gets specialized by domain context (Trans,LGBT) narrowing the semantic scope.*
[DNA, genetic]	[mod, mods]	Activists are opening up a new front in their campaign against genetic modification. The latest target is genetically-**mod**ified trees	[modified, edit]	*Multiple modifiers stack to create specific compound meanings. Interaction binds genetic with the action modification.*
[secret, hidden]	[Snowden, WikiLeaks]	On May 24th PBS aired a Frontline documentary about alleged Wikileaker Bradley Manning called "**WikiSecrets**"	[secret, secrets]	*Feature interaction binds topical concepts to create coined term that could not be modeled via co-occurrence alone.*

Table 4. **Third-Order Interaction Examples Captured by PolySAE.** Cubic interactions condition pairwise compositions on additional context, disambiguating meaning through three-way binding. Vanilla SAE typically activates broader or less specific features.

Poly F_1	**Poly** F_2	**Poly** F_3	**Context**	**SAE**	**Observed pattern**
[proved, proven]	[star, stars, superstar]	[reputation, fame]	David Bowie proved some **stars** are big enough not to make themselves available	[star, stars, superstar]	*Three-way relational binding, all arguments must be present; reputation disambiguates which aspect of stars is relevant to the proving action.*
[nuclear, reactor]	[test, testing]	[radiation, magnetic]	US tests **nuclear**-capable missile with the range to strike North Korea	[nuclear, atomic]	*Specifying concept; Event type (testing) × domain (nuclear) × capability (radiation) (Parsons, 1990)*
[black, racial]	[Americans, Canadians]	[people, women]	In a push to get more Black **Americans** involved in the world of tech	[Americans, Muslims, Jews]	*Multi-attribute category intersection, binding demographic attributes.*
[ing, ting]	[stock, market]	[invest, investing]	**Invest**ing.com — Philippines stocks were higher after	[ing, training, running]	*Three-way interaction between morphological marker (ing) and domain (stock, market) (Asher, 2011).*
[historic, historical]	[UFC, MMA]	[strong, impressive]	Jon Jones' **historic** UFC title reign came to an end	[the, ,, .]	*The standard for historic is calibrated by specific domain and assessed quality; Degree (historic) × domain (UFC) × evaluation (strong, impressive)*

generic entities such as *nuclear* or *Americans* based on surrounding semantic attributes. Across examples, **higher-order terms absorb contextual variation that would otherwise fragment linear features**, allowing PolySAE to express compositional meaning through structured interactions rather than proliferating context-specific atoms. Further examples in Section C confirm these patterns.

6. Conclusion

We introduced PolySAE, a sparse autoencoder that extends the decoder with higher order terms to model feature interactions while preserving a linear encoder for interpretability. Through low-rank tensor factorization on a shared projection subspace, PolySAE captures pairwise and triple feature interactions with small parameter overhead. Across four language models and three SAE variants, PolySAE consistently improves probing F1 by 8% on average while maintaining comparable reconstruction error. PolySAE also achieves 2–10× larger Wasserstein distances between class-conditional feature distributions, indicating that polynomial decoding produces representations with more separated semantic structure. Crucially, learned interaction weights exhibit negligible correlation with co-occurrence frequency ($r = 0.06$), suggesting that the model allocates interaction capacity based on compositional structure rather than sur-

face statistics. **Limitations:** We study models up to 2B parameters and, despite general applicability, restrict experiments to forced-sparsity SAE variants.

Impact Statement

This work contributes to the field of interpretable machine learning by introducing PolySAE, a method for modeling non-additive feature interactions in sparse autoencoders. By enabling explicit representation of compositional structure while preserving linear, human-interpretable features, this approach advances tools for mechanistic analysis of large language models. Improved interpretability has the potential to support downstream efforts in model auditing, debugging, and safety research by making it easier to identify, analyze, and intervene on meaningful internal representations.

The primary anticipated benefits of this work are methodological and scientific. PolySAE is intended as an analysis tool rather than a deployment-facing component, and it does not directly increase the capabilities of language models. As with other interpretability methods, there is a possibility that insights into internal representations could be misused to more effectively manipulate model behavior, but we do not identify novel or unique risks introduced by this work beyond those already present in the interpretability literature.

Overall, we believe this work has a net positive societal impact by strengthening the technical foundations of interpretability and contributing to the long-term goal of building more transparent, controllable, and trustworthy machine learning systems.

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